Kisan AI — Frontend, Backend & Core Features

1. Frontend

Kisan AI frontend must be **mobile-first** because farmers will mainly use it on their phones in the field.

Dashboard

The dashboard should have simple feature cards with **relevant agriculture images**, not only icons and text.

Core cards:

- 🌿 Crop Disease Detection
- 🐛 Pest & Insect Detection
- 💊 Pesticide Recommendation
- 🧴 Insecticide Recommendation
- 🌱 Crop Recommendation
- 🌦️ Weather
- 💧 Irrigation Guide
- 🤖 Ask Kisan AI

Each card should have:

- Relevant agriculture image
- Feature name
- Short description
- Clear button

Mobile Navigation

On mobile, use a **hamburger menu**.

The menu should contain all major options and nothing should be hidden, cut off, or inaccessible.

Crop Image Upload

The farmer must have two options:

Take Photo

- Open phone camera.
- Use the rear camera where possible.

Choose from Gallery

- Select an existing image.

Flow:

Camera/Gallery → Image Preview → Analyze → Result

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2. Backend

Backend should provide APIs for all Kisan AI features.

The backend should handle:

- User authentication
- User profiles
- Crop information
- Image upload
- AI image analysis
- Disease detection
- Pest/insect detection
- Pesticide recommendations
- Insecticide recommendations
- Crop recommendations
- Weather data
- Irrigation recommendations
- AI farming assistant
- Analysis history

The frontend should communicate with the backend through clean REST APIs.

Do not put private API keys or AI credentials in the frontend.

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3. Core Feature — Crop Disease Detection

Farmer uploads a crop/plant image.

Backend sends the image to the AI model.

AI should identify:

- Crop
- Possible disease

- Disease confidence
- Visible symptoms
- Possible causes
- Recommended treatment
- Prevention tips

The result should be easy for a farmer to understand.

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4. Core Feature — Pest & Insect Detection

This is a separate core feature.

Farmer can take/upload a photo of:

- Insect
- Pest
- Damaged leaf
- Damaged fruit
- Crop damage

AI should analyze the image and provide:

- Possible pest/insect name
- Confidence
- Visible damage
- Affected crop
- Possible cause
- Recommended control method

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5. Core Feature — Pesticide Recommendation

Based on:

- Crop
- Disease
- Pest
- Symptoms
- AI analysis
- Farmer's location
- Crop growth stage where available

Backend should provide suitable pesticide guidance.

The result should show:

- Problem
- Recommended pesticide type/product category
- Why it is recommended
- Basic application guidance
- Safety precautions
- Important warnings

Do not blindly recommend chemicals without considering the detected problem.

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6. Core Feature — Insecticide Recommendation

For insect/pest problems, provide insecticide guidance.

Input can come from:

- AI pest detection
- Farmer's selected crop
- Farmer's description
- Uploaded image

Result should include:

- Detected/selected pest
- Recommended insecticide type
- Why it is recommended
- Basic usage guidance
- Safety precautions
- Prevention advice

The system should clearly separate:

Disease → Pesticide/Fungicide guidance

from

Insect/Pest → Insecticide guidance

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7. Crop Recommendation

Farmer provides information such as:

- Location
- Season
- Soil type
- Water availability
- Farm conditions

Backend returns suitable crops with simple reasons.

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8. Weather

Backend should fetch weather data based on the farmer's location.

Frontend should show:

- Temperature
- Weather condition
- Rain probability
- Humidity
- Wind
- Forecast

Weather information can also be used by other features such as irrigation recommendations.

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9. Irrigation Guide

Backend can use:

- Crop
- Weather
- Temperature
- Rain forecast
- Soil information
- Crop stage

to provide simple irrigation guidance.

Example:

> "Rain is expected tomorrow. Irrigation may not be necessary today."

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10. Ask Kisan AI

Provide a simple AI chat.

Farmer can ask questions about:

- Crops
- Diseases
- Pests
- Insects
- Pesticides
- Insecticides
- Fertilizers
- Irrigation
- Weather
- General farming

Support:

- Urdu
- Roman Urdu
- English

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11. Backend Architecture

Backend should be modular.

Suggested modules:

```
Auth
Users
Crops
Disease Detection
Pest Detection
Pesticide Recommendations
Insecticide Recommendations
Weather
Irrigation
AI Assistant
Image Upload
History
```

AI services should be separated from normal API/business logic.

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12. Important Rule

Kisan AI should not only say:

> "This is Disease X."

It should answer the farmer's actual question:

> **What is the problem?**
> **Why did it happen?**
> **What should I do now?**
> **What should I avoid?**

The same approach should be used for pests, pesticides, and insecticides.

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13. Main Farmer Flow

```
Login
  ↓
Dashboard
  ↓
Select Feature
  ↓
Take Photo / Gallery
  ↓
AI Analysis
  ↓
Problem Detection
  ↓
Treatment / Pesticide / Insecticide Recommendation
  ↓
Safety & Prevention Advice
```

The complete system should be **simple, fast, mobile-first, and farmer-friendly**.